Probing the Limits of Neural Thicket Ensembles: Modest Adversarial Gains, No Benefit from Structured Perturbation Diversity

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Abstract

We study whether neural thicket ensembles built by perturbing a single pretrained checkpoint improve robustness to adversarial attacks and out-of-distribution (OOD) inputs. Our main question is whether explicit perturbation-direction design helps beyond simple random perturbations. We train a CIFAR-10 ResNet-18 baseline, generate $M = 6$ experts with matched perturbation scale ($\sigma = 0.015$), recalibrate batch normalization statistics, and compare three ensemble variants: random Gaussian directions, orthogonalized directions, and adversarially selected directions. We evaluate clean accuracy, expected calibration error, negative log-likelihood, projected-gradient-descent (PGD) robust accuracy at $\epsilon \in \{2/255, 4/255, 8/255\}$, OOD detection on SVHN, and diversity diagnostics.

Random perturbation ensembling improves strong-attack robustness over a single model at $\epsilon = 8/255$ by $+0.45$ percentage points (bootstrap 95% CI $[+0.15, +0.80]$, $p = 0.004$), but reduces clean accuracy by -1.29 points (95% CI $[-1.72, -0.86]$, $p < 0.001$). Orthogonal perturbations are statistically indistinguishable from random for PGD@8/255 ($+0.10$ points, 95% CI $[-0.20, +0.40]$, $p = 0.599$). Adversarial-direction perturbations produce much higher disagreement but worse calibration and weaker robustness than random. These results show a measurable but modest robustness gain from perturbation ensembling, while indicating that unconstrained diversity is not sufficient for robust generalization.

1 Introduction

Adversarial and distribution-shift failures remain a practical barrier for deploying image classifiers in high-stakes settings. Deep ensembles improve uncertainty quality, but training many independent models is expensive [Lakshminarayanan et al., 2017]. Neural thicket style methods suggest a cheaper alternative: sample diverse experts around one pretrained solution in weight space [Gan and Isola, 2026, Mehrtash et al., 2020].

why is this still unresolved? Existing studies show that diversity can help uncertainty estimation, but the link between perturbation-generated diversity and robustness is not settled under a controlled adversarial + OOD protocol [Xia and Bouganis, 2022, Wei and Liu, 2020]. In particular, it is unclear whether structured perturbation directions are better than random directions when ensemble size and perturbation scale are fixed.

We examine this question using one CIFAR-10 ResNet-18 base model and three perturbation strategies: random, orthogonal, and adversarial-direction perturbations. We evaluate clean accuracy, calibration, PGD robustness at $\epsilon \in \{2/255, 4/255, 8/255\}$, and OOD detection on SVHN. This design isolates the effect of direction strategy while keeping architecture, data, and expert count constant.

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

Our key quantitative result is narrow but clear. Random perturbation ensembling improves PGD@8/255 from 0.05% to 0.50% (absolute $+0.45$ points, 95% CI $[+0.15, +0.80]$), while reducing clean accuracy from 88.28% to 86.99%. Orthogonal directions do not significantly improve over random at PGD@8/255 ($p = 0.599$), and adversarial-direction perturbations degrade calibration and robustness.

Our contributions are:

- We conduct a controlled robustness benchmark for perturbation-based thicket ensembles across clean, adversarial, OOD, and calibration metrics.
- We compare random, orthogonal, and adversarial-direction perturbation strategies under matched ensemble budget ($M = 6$) and perturbation scale ($\sigma = 0.015$).
- We provide bootstrap confidence intervals and hypothesis tests for primary robustness claims, including negative findings.
- We show that larger disagreement alone does not predict better robustness or OOD separation in this setup.

Paper organization. section 2 reviews prior work. section 3 describes datasets, models, and protocol. section 4 presents quantitative results and statistical tests. section 5 discusses interpretation and limitations, and section 6 concludes.

2 Related Work

perturbation and thicket ensembles. Parameter-space perturbation ensembles provide a compute-efficient alternative to training independent models. PEP showed that perturbing a reference model can recover part of deep-ensemble benefits at lower cost [Mehrtash et al., 2020]. Neural Thickets extended this direction by arguing that many useful experts are dense around pretrained weights [Gan and Isola, 2026]. Our work is aligned with this line, but we focus specifically on robustness stress tests rather than general task transfer.

adversarial robustness in ensembles. Ensemble-based defenses have a long history in adversarial machine learning. Ensemble adversarial training improves transfer robustness by exposing models to attacks from multiple sources [Tramèr et al., 2018]. Certification work also shows that ensemble aggregation can strengthen formal guarantees under some conditions [Ahmad and Evans, 2020, Zhang et al., 2019]. Unlike these approaches, we do not retrain robust models or optimize certification objectives; we isolate post-hoc perturbation direction effects around one pretrained checkpoint.

OOD uncertainty and diversity. Deep ensembles are strong uncertainty baselines for calibration and OOD detection [Lakshminarayanan et al., 2017]. Later OOD studies emphasized architecture and training protocol choices, including leave-out designs and shared-parameter ensembles [Vyas et al., 2018, Chen et al., 2022]. Direct analysis of diversity metrics shows that more disagreement is not always better for OOD detection [Xia and Bouganis, 2022]. Our findings are consistent with this cautionary view: high disagreement from adversarial-direction perturbations did not translate into better OOD AUROC.

positioning. Compared with prior work, our contribution is a controlled four-way comparison `single`, `random`, `orthogonal`, and `adversarial` under matched model, data, and compute budget, with bootstrap inference for key robustness claims.

3 Methodology

Problem setup. We study whether perturbation-generated ensembles around one pretrained model improve robustness to adversarial and OOD shifts. Let $f_{\boldsymbol{\theta}}(\boldsymbol{x})$ be a classifier with parameters $\boldsymbol{\theta}$. For an ensemble with M experts, we construct perturbed parameters

$$\boldsymbol{\theta}_m = \boldsymbol{\theta}_0 + \sigma \boldsymbol{d}_m, \quad m = 1, \ldots, M, \tag{1}$$

where $\boldsymbol{\theta}_0$ is the pretrained checkpoint, $\sigma = 0.015$, and $\boldsymbol{d}_m$ is a direction vector defined by one of three strategies. Ensemble prediction uses probability averaging,

$$p(y \mid \boldsymbol{x}) = \frac{1}{M} \sum_{m=1}^{M} p_m(y \mid \boldsymbol{x}). \tag{2}$$

Method	Clean Acc	ECE ↓	NLL ↓	OOD AUROC	OOD AUPR	FPR95 ↓	PGD@2/255	PGD@4/255	PGD@8/255
Single	**0.8828**	**0.0350**	**0.3821**	0.8389	0.7680	**0.7827**	**0.1375**	0.0055	0.0005
Random	0.8699	0.0360	0.4138	**0.8430**	**0.7705**	0.7891	0.1195	**0.0175**	0.0050
Orthogonal	0.8702	0.0362	0.4129	0.8419	0.7663	0.7855	0.1190	**0.0175**	**0.0060**
Adversarial	0.8402	0.0894	0.5429	0.8228	0.7627	0.8468	0.1135	0.0045	0.0000

Table 1: Main evaluation metrics on CIFAR-10 (ID), SVHN (OOD), and PGD attacks. Best values are shown in bold for each metric. Higher is better except columns marked with ↓.

Datasets and preprocessing. In-distribution (ID) data is CIFAR-10 with 50,000 training and 10,000 test images. OOD data is SVHN (cropped digits) test split with 26,032 images. CIFAR-10 training uses random crop (padding 4) and horizontal flip. All evaluation sets use tensor conversion only, with pixel values in $[0, 1]$ and no dataset mean/std normalization.

Base model and training. We use a CIFAR-adapted ResNet-18 (3×3 stem, no maxpool). Training uses SGD with momentum 0.9, weight decay 5×10^{-4}, initial learning rate 0.1 with cosine decay, label smoothing 0.05, batch size 128, and 8 epochs. Mixed precision is enabled. Global seed is 42.

Ensemble construction. We compare four methods:

- **Single**: the unperturbed baseline model.
- **Random**: Gaussian random normalized directions.
- **Orthogonal**: random directions orthogonalized with Gram–Schmidt in flattened weight space.
- **Adversarial**: directions derived from per-batch loss gradients, then orthogonalized.

All ensemble methods use $M = 6$ experts and per-expert batch-normalization recalibration.

Evaluation protocol. We report clean accuracy, expected calibration error (ECE, 15 bins), and negative log-likelihood (NLL) on CIFAR-10 test. Adversarial robustness is measured by ℓ_∞ PGD with 10 steps at $\epsilon \in \{2/255, 4/255, 8/255\}$. For runtime control, robust evaluation uses the first 2,000 CIFAR-10 test samples. OOD detection compares CIFAR-10 test (ID positive) vs SVHN test using max-softmax confidence, and reports AUROC, AUPR, and FPR95.

Statistical analysis. We use paired bootstrap (3,000 resamples) for primary method differences and report 95% confidence intervals with p-values. We also compute pairwise disagreement and pairwise error correlation among experts, and evaluate Spearman correlation between disagreement and robustness/OOD metrics.

4 Results

Overall performance. table 1 reports clean, calibration, OOD, and adversarial metrics for all methods. The single model has the best clean accuracy (0.8828), ECE (0.0350), and NLL (0.3821). Random and orthogonal perturbation ensembles improve strong-attack robustness at PGD@8/255 (0.0050 and 0.0060) relative to the single model (0.0005), but both reduce clean accuracy by about 1.3 points.

Statistical tests for primary hypotheses. table 2 shows bootstrap inference for predefined comparisons. Random perturbation ensembling improves PGD@8/255 over single by $+0.0045$ (95% CI $[+0.0015, +0.0080]$, $p = 0.004$), supporting a limited version of H1 at high attack strength. Orthogonal directions are not better than random ($p = 0.599$), so H2 is not supported. Adversarial-direction perturbations are worse than random (difference -0.0050, $p < 0.001$), refuting H3 in this setting.

Diversity does not imply robustness. table 3 shows that adversarial-direction perturbations produce much larger disagreement (0.2125) and much lower error correlation (0.4922) than random or orthogonal methods. However, this diversity increase does not improve robustness or OOD detection. Spearman analysis also fails to show positive association between disagreement and outcomes (disagreement vs OOD AUROC: $\rho = -0.20$, $p = 0.8$; disagreement vs PGD@8/255: $\rho = -0.40$, $p = 0.6$), so H4 is not supported.

Runtime and reproducibility. Full evaluation after checkpoint load takes 5.68 minutes on two RTX 3090 GPUs (24GB each), with fixed method seeds {42, 101, 202, 303}. This makes perturbation

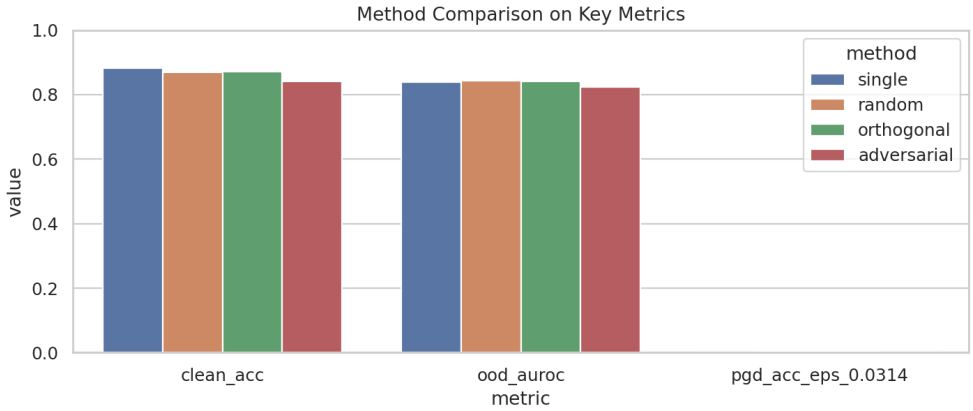

Figure 1: Comparison of key metrics across methods. Perturbation ensembles shift the clean–robustness trade-off: small gains under stronger attack come with clean-performance degradation.

Comparison (difference)	Estimate	95% CI	p-value
PGD@8/255: Random − Single	+0.0045	[+0.0015, +0.0080]	0.004
PGD@8/255: Orthogonal − Random	+0.0010	[-0.0020, +0.0040]	0.599
PGD@8/255: Adversarial − Random	-0.0050	[-0.0085, -0.0020]	< 0.001
Clean Acc: Random − Single	-0.0129	[-0.0172, -0.0086]	< 0.001

Table 2: Bootstrap results for primary hypotheses. Differences are reported in absolute proportion units.

ensembling operationally lightweight, but the absolute robust accuracies remain very low ($< 1\%$ at PGD@8/255) for all non-adversarially-trained methods.

5 Discussion

Interpreting the main outcome. The results support a narrow claim: perturbation ensembling can improve robustness under stronger attack, but the gain is modest and does not generalize across all threat levels. At PGD@8/255, random perturbations help relative to a single model, yet all methods remain near-collapse in absolute terms. This indicates that post-hoc ensembling alone is not a substitute for robust training.

what did structured diversity miss? Orthogonal and adversarial direction design did not outperform random perturbations under fixed $\sigma = 0.015$ and $M = 6$. The adversarial-direction method produced very high disagreement but worse calibration and lower clean accuracy. This suggests that diversity must be constrained by local predictive quality; maximizing disagreement without such constraints can move experts into poorly calibrated regions.

Practical implications. For teams that already have a strong pretrained checkpoint, random perturbation ensembling is a low-effort option that can produce small robustness gains under severe attack. However, the clean-accuracy penalty and weak absolute robust accuracy mean this approach should be treated as a supplementary defense, not a primary safeguard.

Limitations. Our study uses one base-training seed and one perturbation scale in the final comparison, evaluates PGD on a 2,000-sample subset for runtime control, and uses only SVHN as OOD data. We also do not include AutoAttack or adversarially trained baselines in the final table. These constraints limit external validity and likely underestimate uncertainty across training runs.

Broader implications. The central lesson is methodological: useful diversity is not equivalent to large diversity. Robustness-oriented expert generation likely requires explicit constraints or objec-

Method	Pairwise Disagreement	Pairwise Error Correlation
Random	0.00619	0.9788
Orthogonal	0.00544	0.9811
Adversarial	**0.2125**	**0.4922**

Table 3: Diversity diagnostics among ensemble experts. Higher disagreement and lower error correlation indicate stronger diversity.

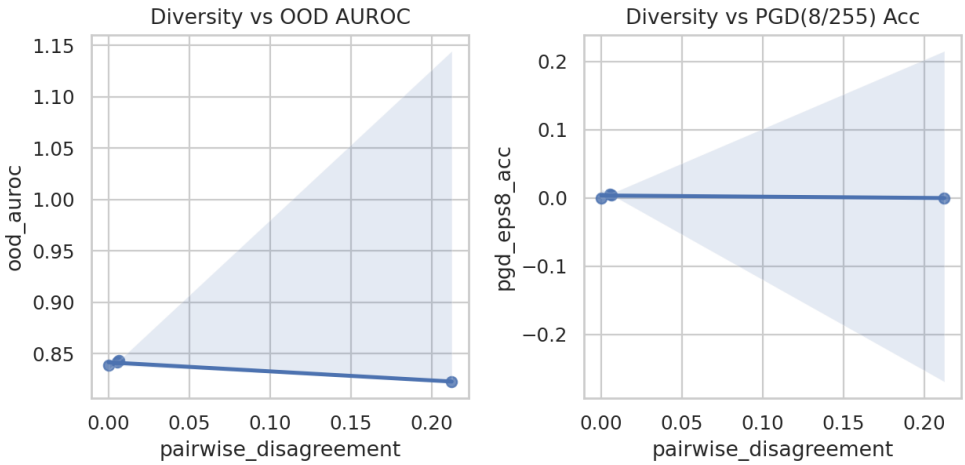

Figure 2: Correlation diagnostics between diversity and robustness/OOD metrics across methods. Larger disagreement does not map to better robustness in this experiment.

tives that preserve clean decision quality while spreading gradients and errors in meaningful directions.

6 Conclusion

We presented a controlled benchmark of perturbation-based neural thicket ensembles for adversarial and OOD robustness on CIFAR-10/SVHN. Under matched architecture and budget, random perturbation ensembling provided a statistically significant but modest gain at the strongest attack level tested (PGD@8/255), while reducing clean accuracy. Orthogonal direction design did not significantly improve over random, and adversarial-direction perturbations degraded calibration and robustness despite much higher disagreement.

The main takeaway is simple: perturbation ensembling can help at the margin, but diversity magnitude alone is not a reliable path to robust performance. Future work should combine perturbation ensembling with robust base training, sweep perturbation scale and ensemble size, and optimize direction generation with explicit clean-calibration constraints and robustness-aware objectives.

References

Aounon Kumar Ahmad and David Evans. Certifying joint adversarial robustness for model ensembles. *arXiv preprint arXiv:2004.10250*, 2020.

Jinsol Chen et al. Batch-ensemble stochastic neural networks for out-of-distribution detection. *arXiv preprint arXiv:2206.12911*, 2022.

Yulu Gan and Phillip Isola. Neural thickets: Diverse task experts are dense around pretrained weights. *arXiv preprint arXiv:2603.12228*, 2026.

Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Alireza Mehrtash et al. Pep: Parameter ensembling by perturbation. *arXiv preprint arXiv:2010.12721*, 2020.

Florian Tramèr, Alexey Kurakin, Nicolas Papernot, Ian Goodfellow, Dan Boneh, and Patrick McDaniel. Ensemble adversarial training: Attacks and defenses. In *International Conference on Learning Representations*, 2018.

Apoorv Vyas, Nithin Jammalamadaka, Xu Zhu, Dipankar Das, Bharath Kaul, and Theodore Willke. Out-of-distribution detection using an ensemble of self supervised leave-out classifiers. *arXiv preprint arXiv:1809.03576*, 2018.

Wenqi Wei and Ling Liu. Robust deep learning ensemble against deception. *arXiv preprint arXiv:2009.06589*, 2020.

Yifei Xia and Christos-Savvas Bouganis. On the usefulness of deep ensemble diversity for out-of-distribution detection. *arXiv preprint arXiv:2207.07517*, 2022.

Huan Zhang et al. Enhancing certifiable robustness via a deep model ensemble. *arXiv preprint arXiv:1910.14655*, 2019.